Ashruta Acharya

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Economics Ph.D. candidate specializing in observational causal inference and large-scale data analysis. Builds original datasets and reproducible Python, R, and Stata workflows by integrating decades of census and geospatial data to study infrastructure, market integration, and employment. Seeking economist and quantitative research roles in industry.

Education

George Mason University - Ph.D. in Economics Expected May 2027

Dissertation committee: Mark Koyama (Advisor), Alex Tabarrok, Tyler Cowen, Vincent Geloso.

Fields: Institutions & Development; Economic History

Wichita State University - M.A. in Applied Economics May 2021

Wichita State University - B.B.A. in Economics May 2019

Technical Skills

Methods: Causal inference: difference-in-differences (DiD), event studies, instrumental variables (IV), regression discontinuity (RDD), and synthetic control; panel data, heterogeneous effects, sensitivity analysis.

Software: Python, R, Stata; SQL (working knowledge); ArcGIS Pro, QGIS; Git/GitHub, Excel, LaTeX.

Python libraries: pandas, NumPy, statsmodels, SciPy, GeoPandas, NetworkX, Rasterio, PyArrow.

Data & text: Data integration and validation, spatial joins, boundary harmonization, network accessibility; NLP and AI-assisted document extraction.

Research Experience

Ph.D. Fellow & Research Assistant Aug 2021 - Present

George Mason University *Fairfax, VA*

- Estimated gains in non-agricultural employment following first road connections, with no clear additional gains from subsequent market-access improvements, using staggered difference-in-differences and event studies.
- Built an original four-decade panel of 74 Nepali districts (1971-2011), integrating transport networks and census data and reconciling definitions and boundaries to enable consistent comparisons across regions and time.
- Reduced runtime from over 30 hours to about two hours in a prior market-access analysis, developing reusable Python workflows to process four raster layers with nearly 60 million cells each.
- Documented a roughly 40% decline in the standard deviation of log market access from 1971 to 2011, constructing transport-cost-based measures with destination populations held fixed.
- Extracted people, places, and institutions from 165 legal articles and over 4,000 archival pages using AI-assisted NLP prototypes; manually spot-checked outputs to assess extraction quality.

Job market paper: *Roads, State Integration, and Uneven Development in Modern Nepal.*

Additional Analytical Experience

Research Intern - Institute for Integrated Development Studies (IIDS) Feb-Aug 2021

Center for Economic Policy *Kathmandu, Nepal*

- Produced a product-level profile of Nepal's trade composition and trading-partner concentration, analyzing BACI data spanning 1998-2018.
- Co-authored *Trade Flow Patterns in Nepal*, an IIDS study with financial and technical support from IFPRI; communicated findings through maps, figures, and written summaries.

Graduate Research Assistant Jun 2019 - May 2021

Institute for the Study of Economic Growth, Wichita State University

- Identified 11 Kansas industries with both wage and employment gains by benchmarking 2014-2018 QCEW data against U.S. trends; summarized findings in research memoranda developed with faculty.

Publications

Acharya, A., Geloso, V., & Psurek, A. (2026). *The Hole in the Doughnut: Formalizing and Testing a Key Model of Degrowth.* Contemporary Economic Policy.

Neopane, A., Nepal, B., & Acharya, A. (2021). *Trade Flow Patterns in Nepal.* Feed the Future (USAID), Research Paper 7.

Teaching & Leadership

Graduate Lecturer May 2023 - Present

Department of Economics, George Mason University *Fairfax, VA*

- Explained economic models and empirical evidence to classes of 20-35 students with varied quantitative preparation; designed courses, applied econometrics exercises, and assessments.
- Undergraduate courses: Economics of Developing Areas, Economic History, Intermediate Macroeconomics, Money & Banking, and Economics for the Citizen.

Founding President Jun 2020 - Dec 2021

Society of Nepali Women in Economics (SONWE) *Kathmandu, Nepal*

- Founded a network of 10+ emerging economists and organized Women in Nepali Economic Growth, coordinating mentorship, speakers, and public programming.

Selected Fellowships & Training

Price Theory Summer Camp, Becker Friedman Institute, University of Chicago June 2026

Public Choice Fellowship ($20,000 research support) May-Dec 2026

Graduate Fellow, American Institute for Economic Research Feb-Apr 2024

Mercatus Center Doctoral Fellowship ($160,000 four-year award) 2021-2025

Languages

English (fluent), Nepali (native), Hindi (fluent), Korean (intermediate).